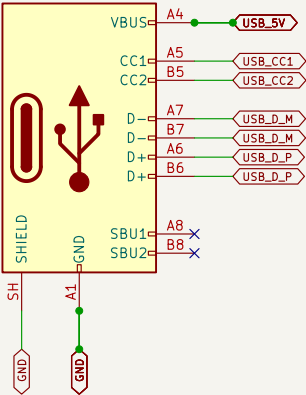


TRIMIX USB INPUT | GCT + ESD + SIX-WIRE HARNESS

Dedicated 0.60 +/-0.10 mm board. Nominal 5 V input; USB-C attachment and BC1.2 detection are on the main PCB.

01 SEALED USB-C RECEPTACLE

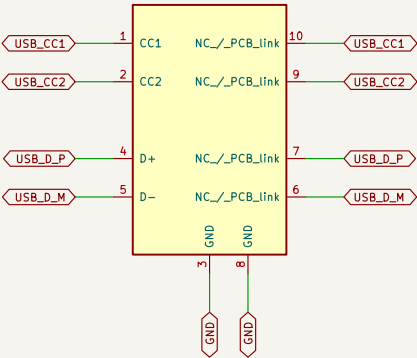
J901
USB4720-03-A



Shield and all ground contacts join GND.
A6/B6 join D+; A7/B7 join D-. SBU is NC.
No local Rd: TUSB320LAI on main PCB supplies Rd.

02 FOUR-LINE + VBUS ESD PROTECTION

U901
TPD4E05U06DQAR



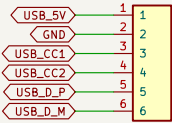
D901
TPD1E10B06DYAR



No internal links on pins6/7/9/10. PCB copper joins
1-10(CC1),2-9(CC2),4-7(D+),5-6(D-).
D901 surge clamp exceeds 6 V; upstream transient
coordination remains a main-system requirement.

03 SIX-WIRE OUTPUT

J902
TO MAIN J101 | SAME PIN NUMBERS



1 = USB_5V 2 = GND
3 = CC1 4 = CC2
5 = D+ 6 = D-
Do not swap CC/data wires or populate old Rd.

04 ASSEMBLY AND QUALIFICATION

GCT Rev B cutout and fixed connector pose retained.
No copper-edge exception rule;0.10mm GCT edge margin needs closure.
Factory stencil/reflow for U901: support the port internally.
Measure harness/solder height inside the 5.7 mm gap.
No 9/12/20 V request: tests required for both cable orientations,
source change, hot plug, overcurrent and assembled-port ESD.

GCT edge margin unresolved; factory assembly and physical tests pending.
Engineering review draft - not released for manufacture

Trimix

Sheet: /
File: Trimix_USB_Input.kicad_sch

Title: Trimix USB input / source and ESD interface

Size: A3 Date: 2026-09-07 Rev: A3-USB6-ESD
KiCad E.D.A. 10.0.6 Id: 1/1